

Connector from gate drivers

Includes multiple redundant SSO (six-switch-off) pathways:

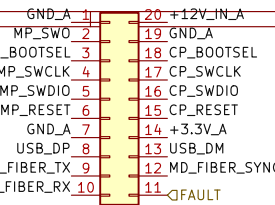
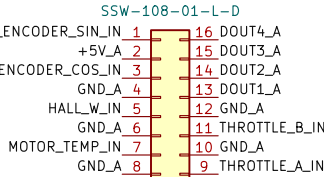
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2

(Gate driver power enable implements 1002, with monitoring feedback).

feedback pins are 0-3.3v binary (3.3v expected when +12v present)

Connector to IO board

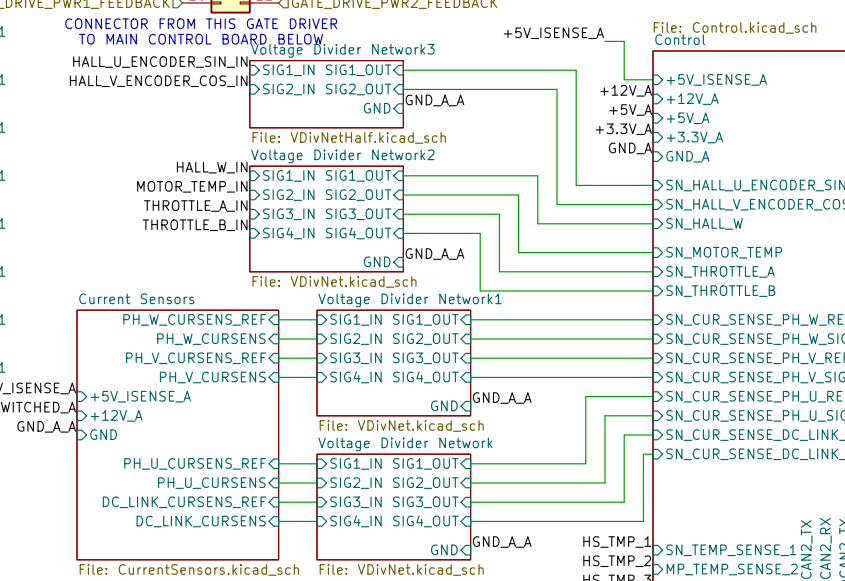
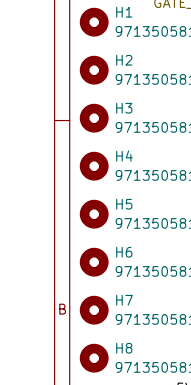
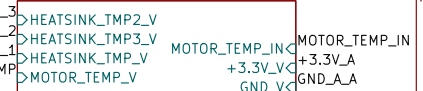
Digital Outputs provided here to IOBoard.



This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)

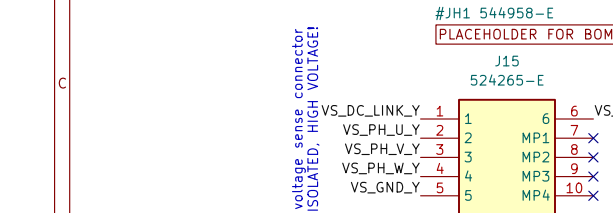
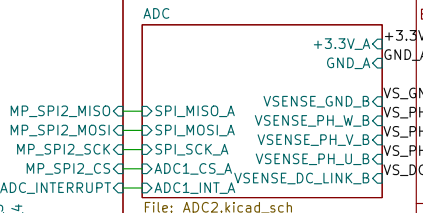
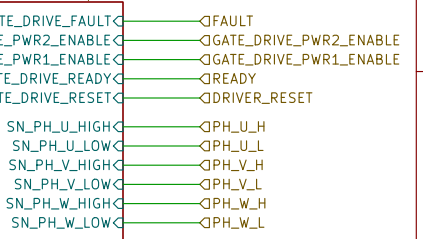
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.

Temperature Sensors



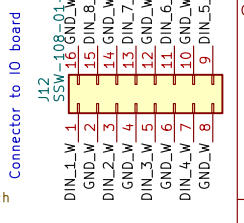
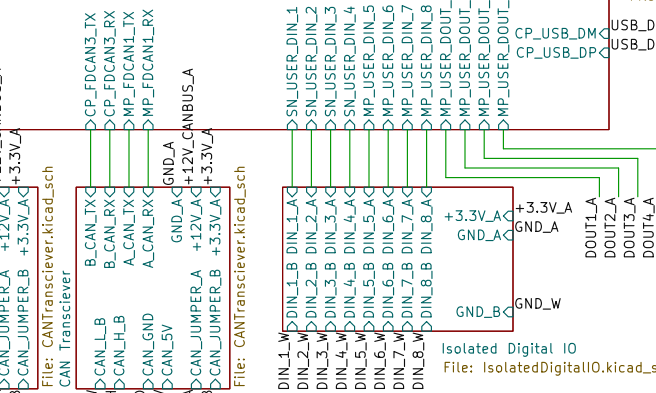
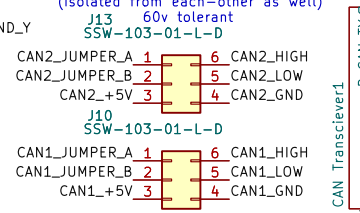
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults

Please see HARA under documentation for additional informaton.

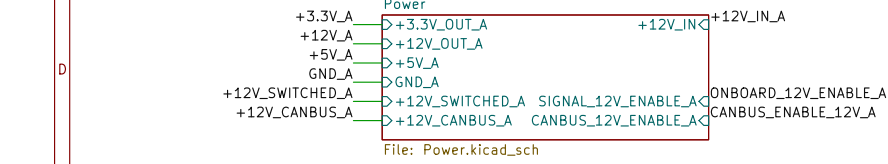


Connector to IO board

Provides dual fully isolated canbus interfaces (isolated from each-other as well)



onboard power regulator section, for 3.3v mcu power, +5v current sense, etc

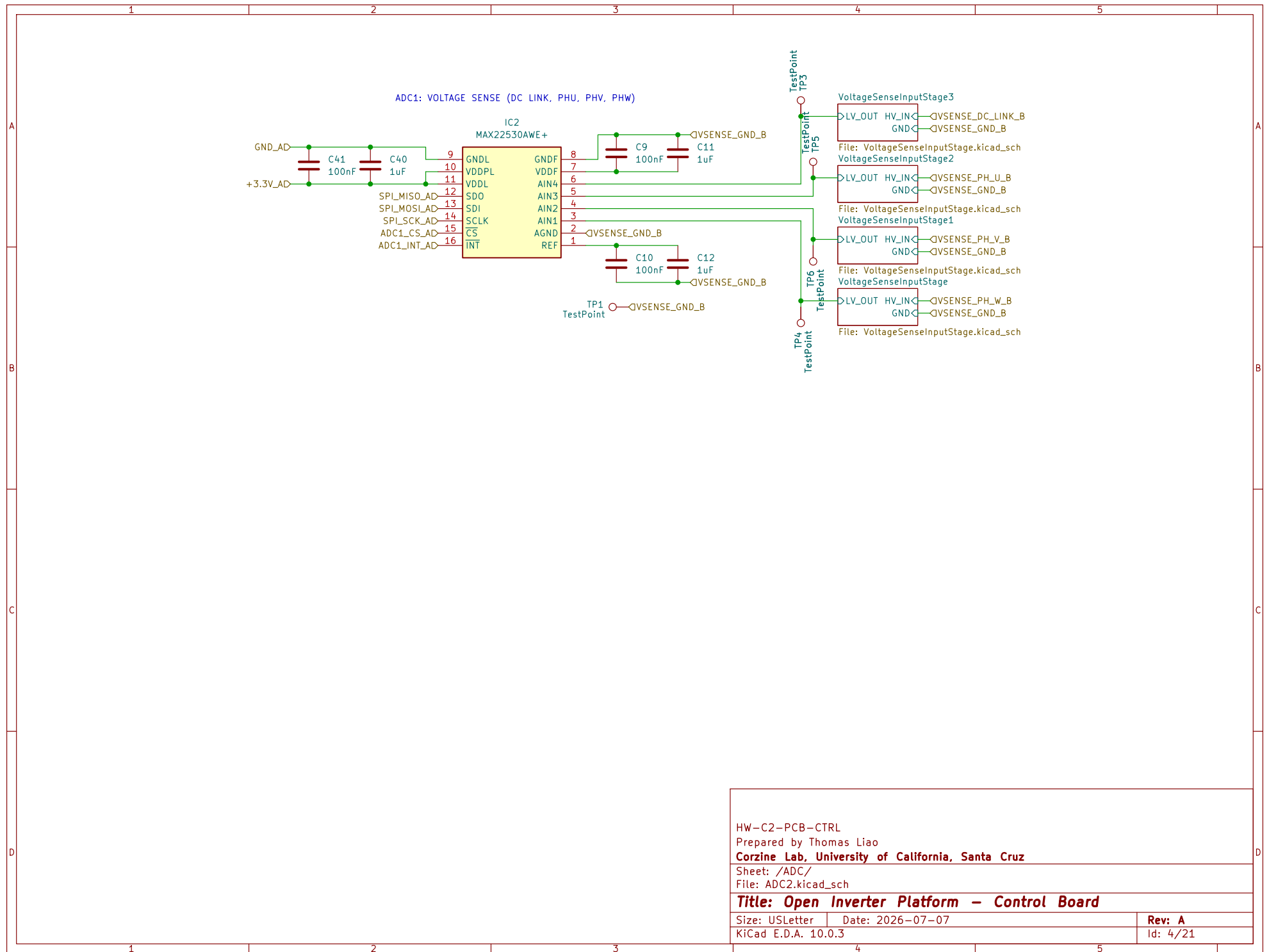


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter Date: 2026-07-07
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Rev: A
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HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

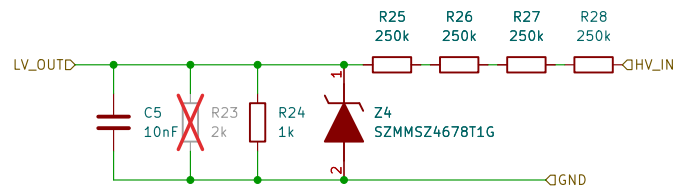
Title: Open Inverter Platform – Control Board

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ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

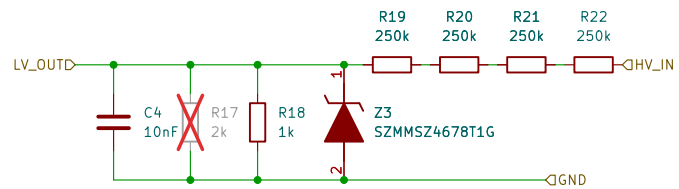
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

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Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

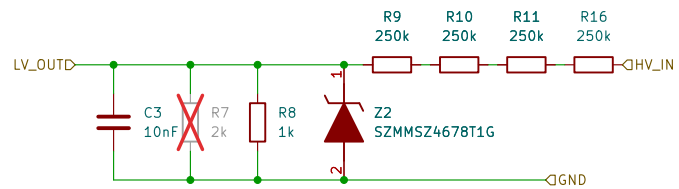
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

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Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

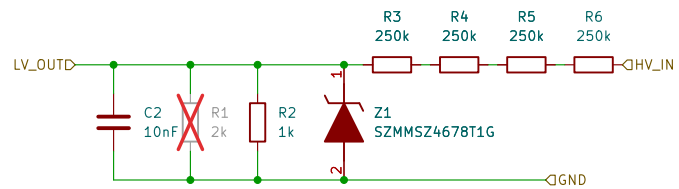
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

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Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

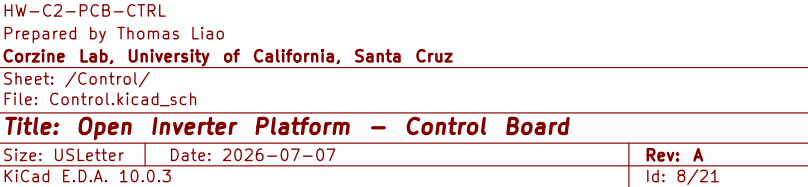
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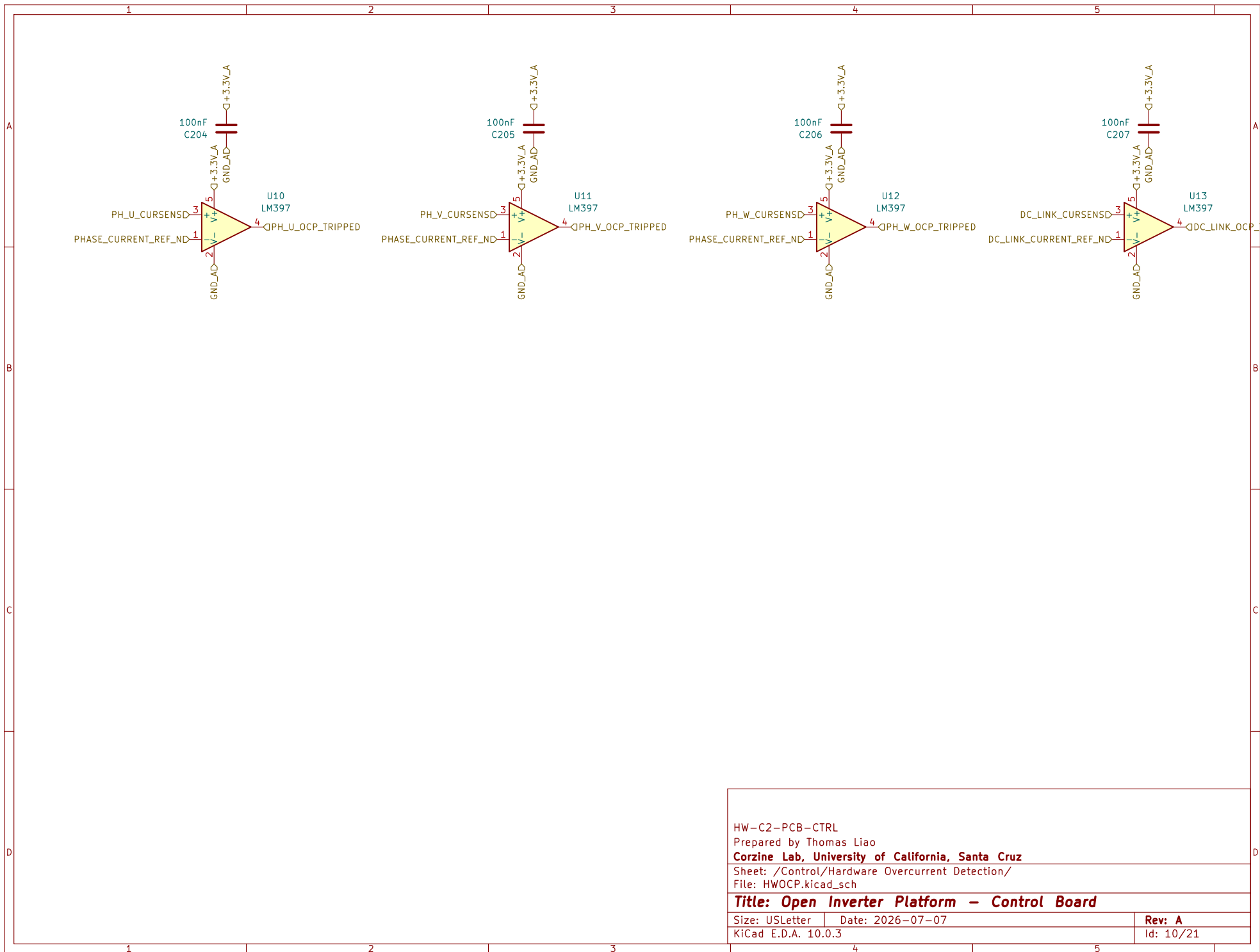
Size: USLetter Date: 2026-07-07

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Rev: A

Id: 6/21





HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

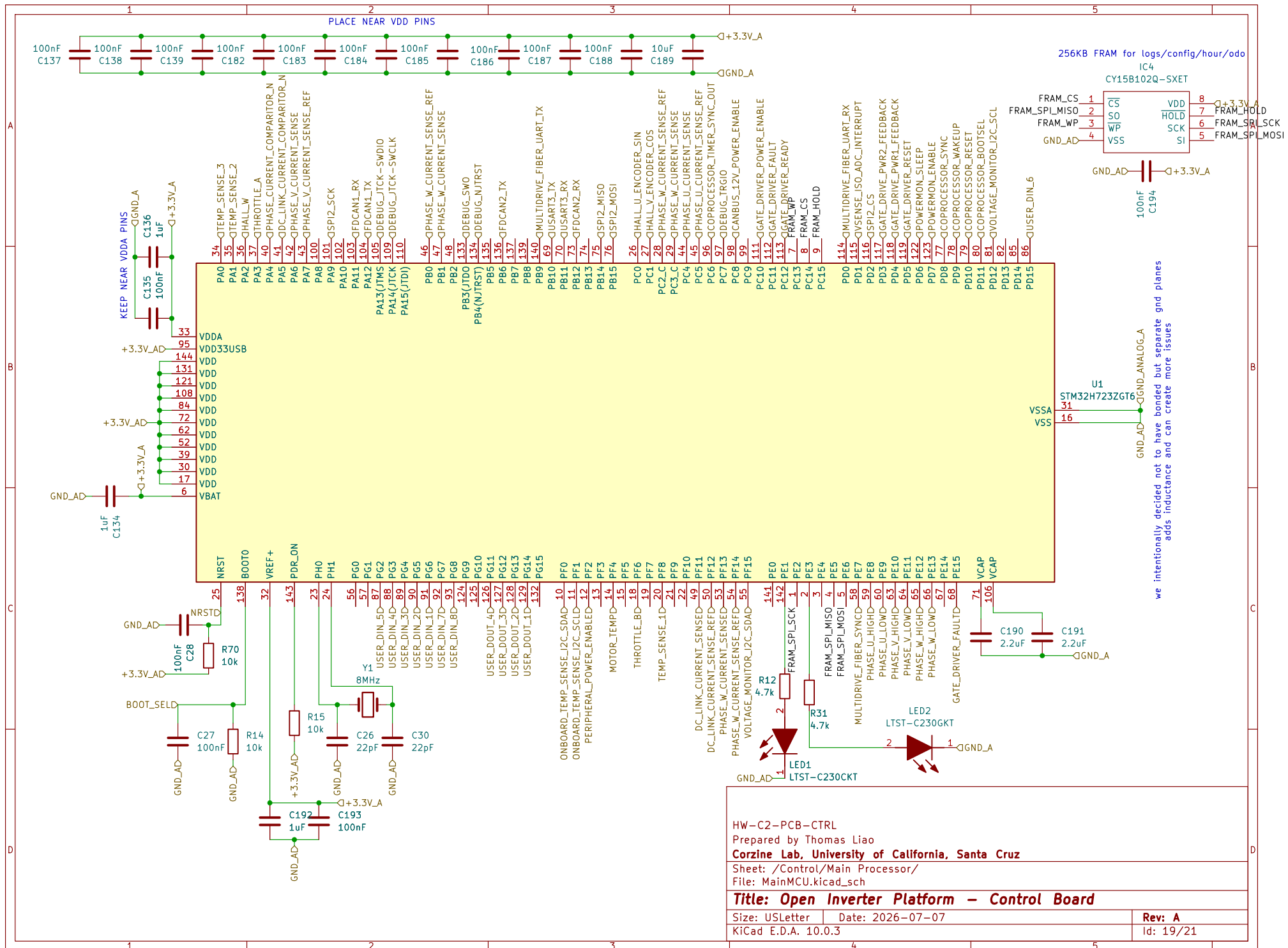
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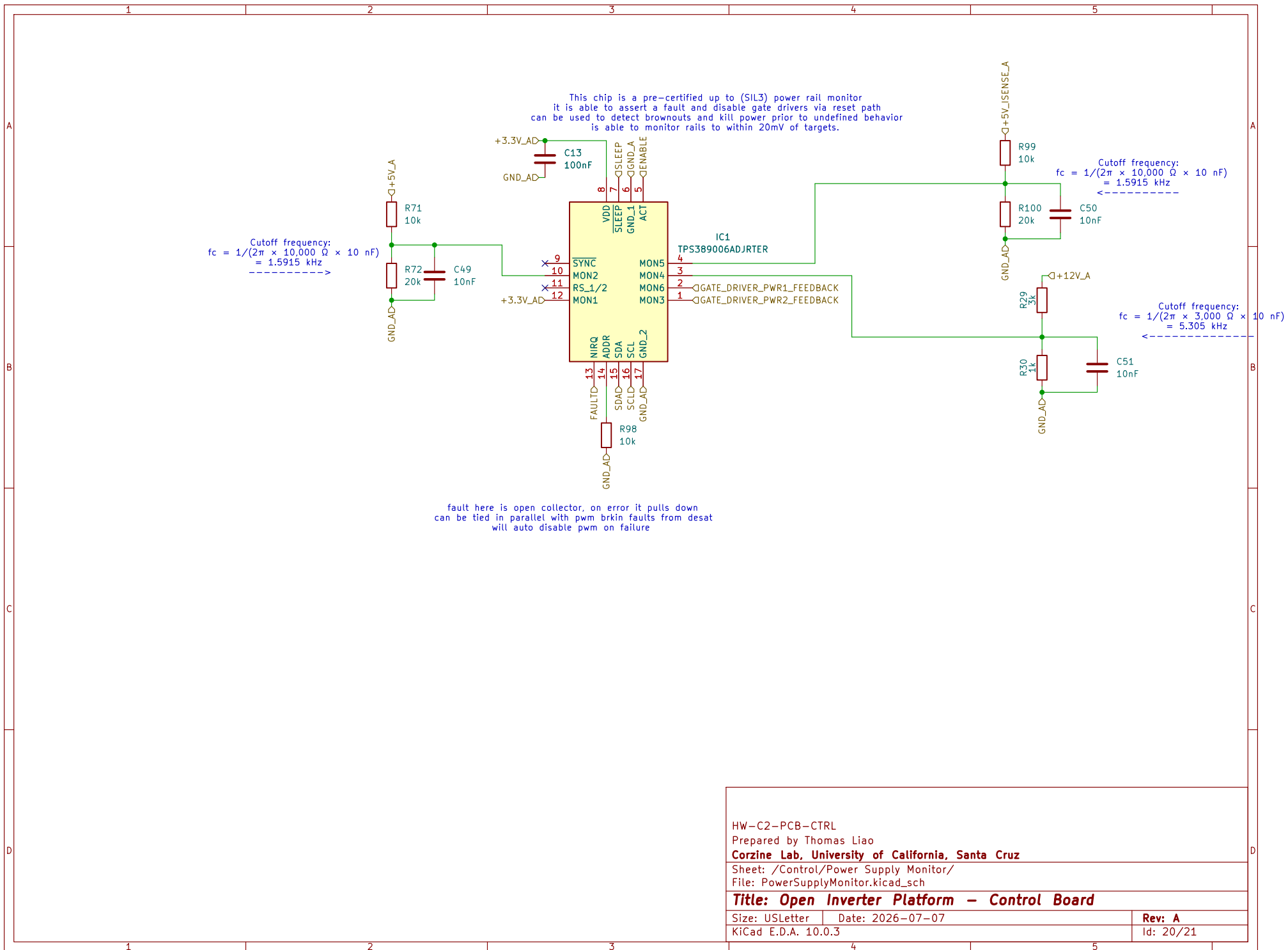
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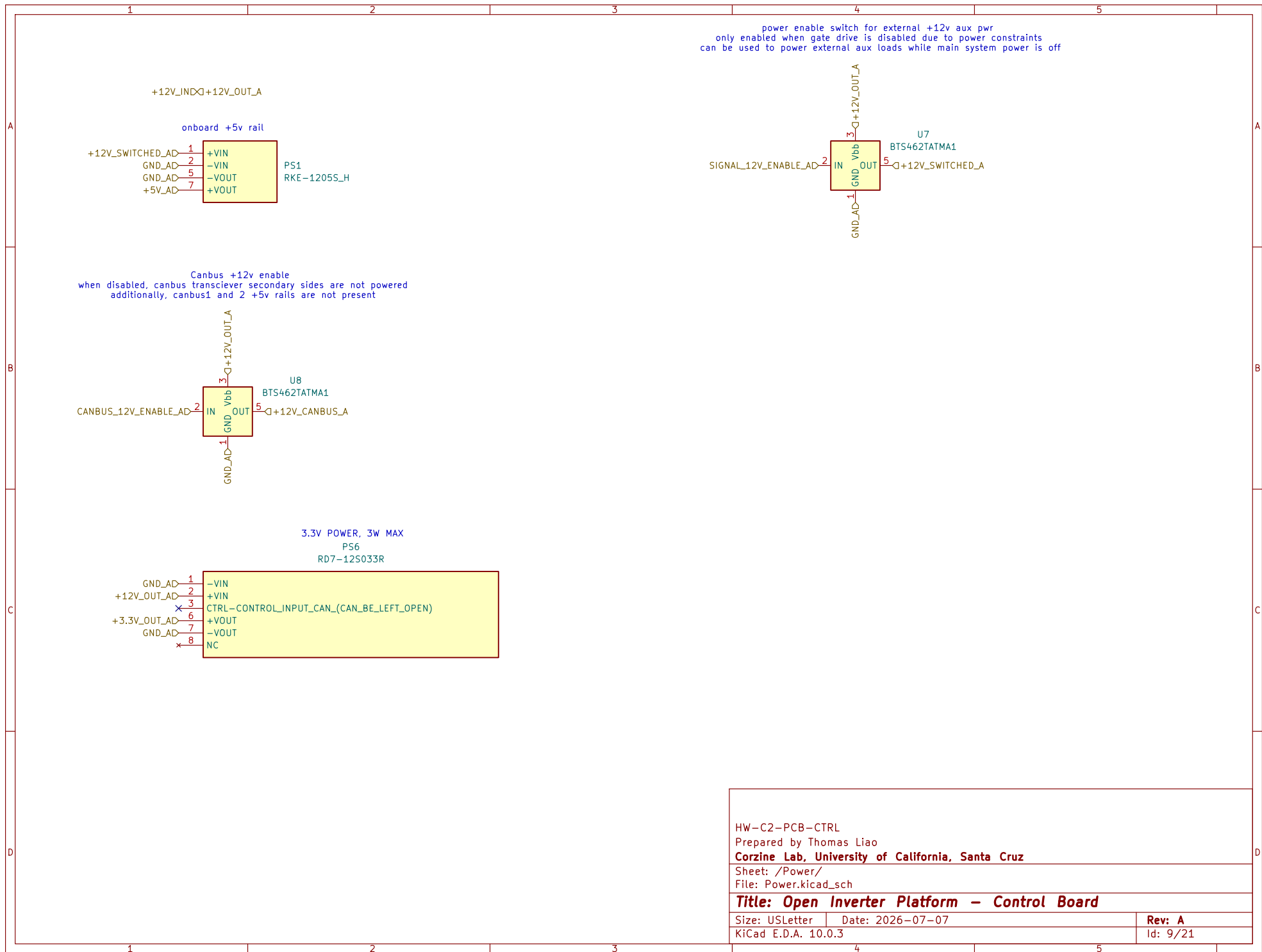
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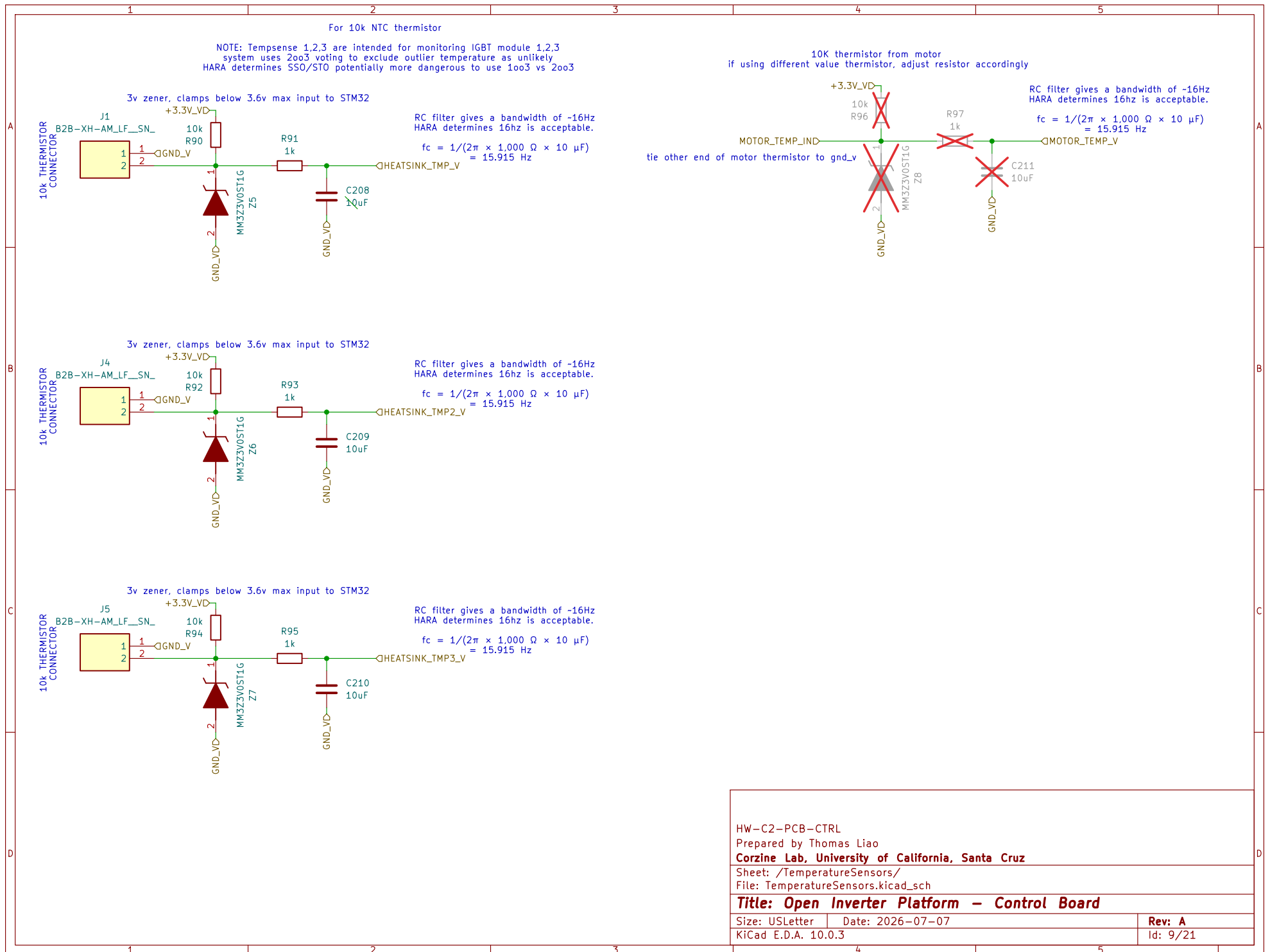
Rev: A

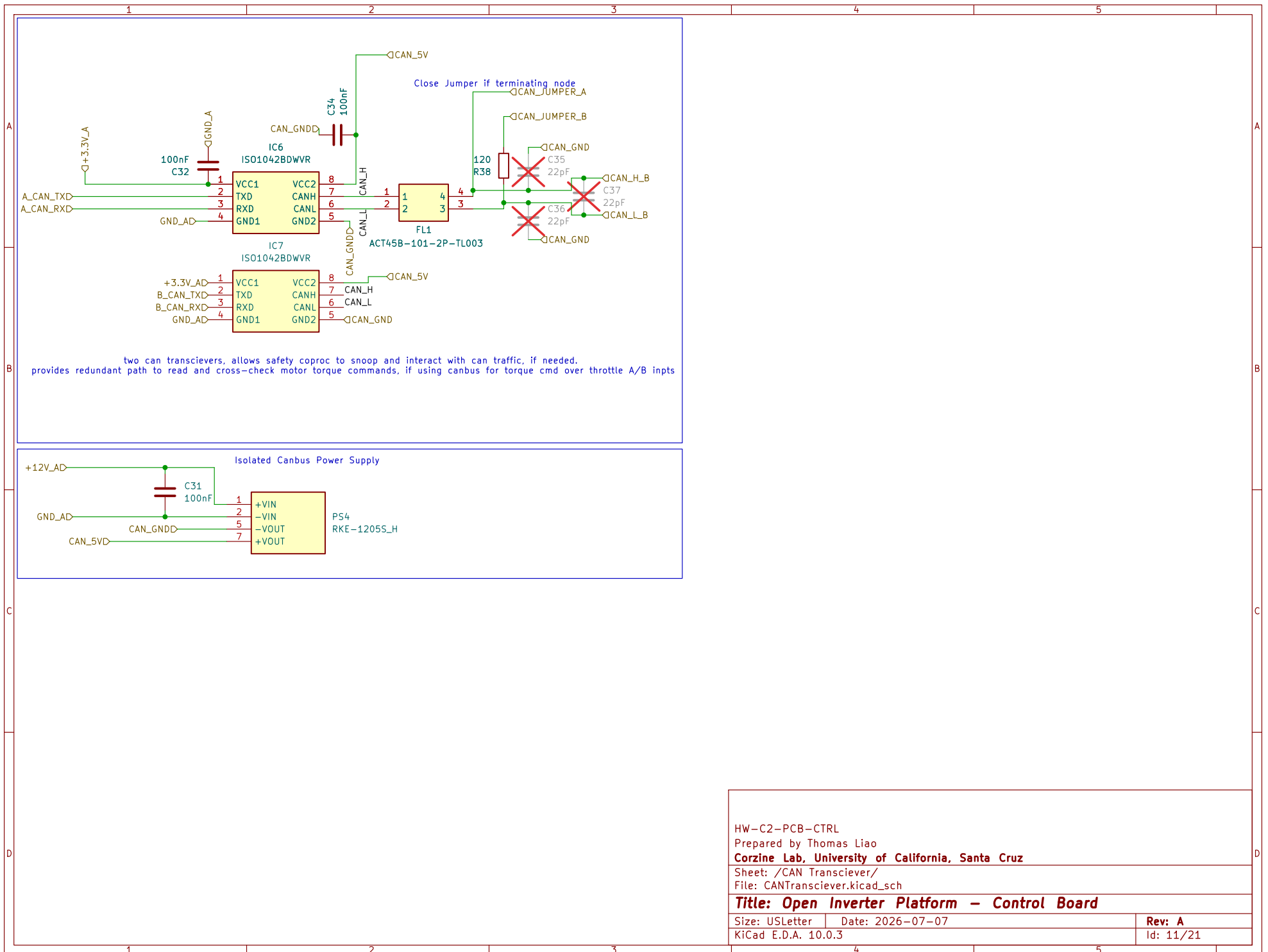
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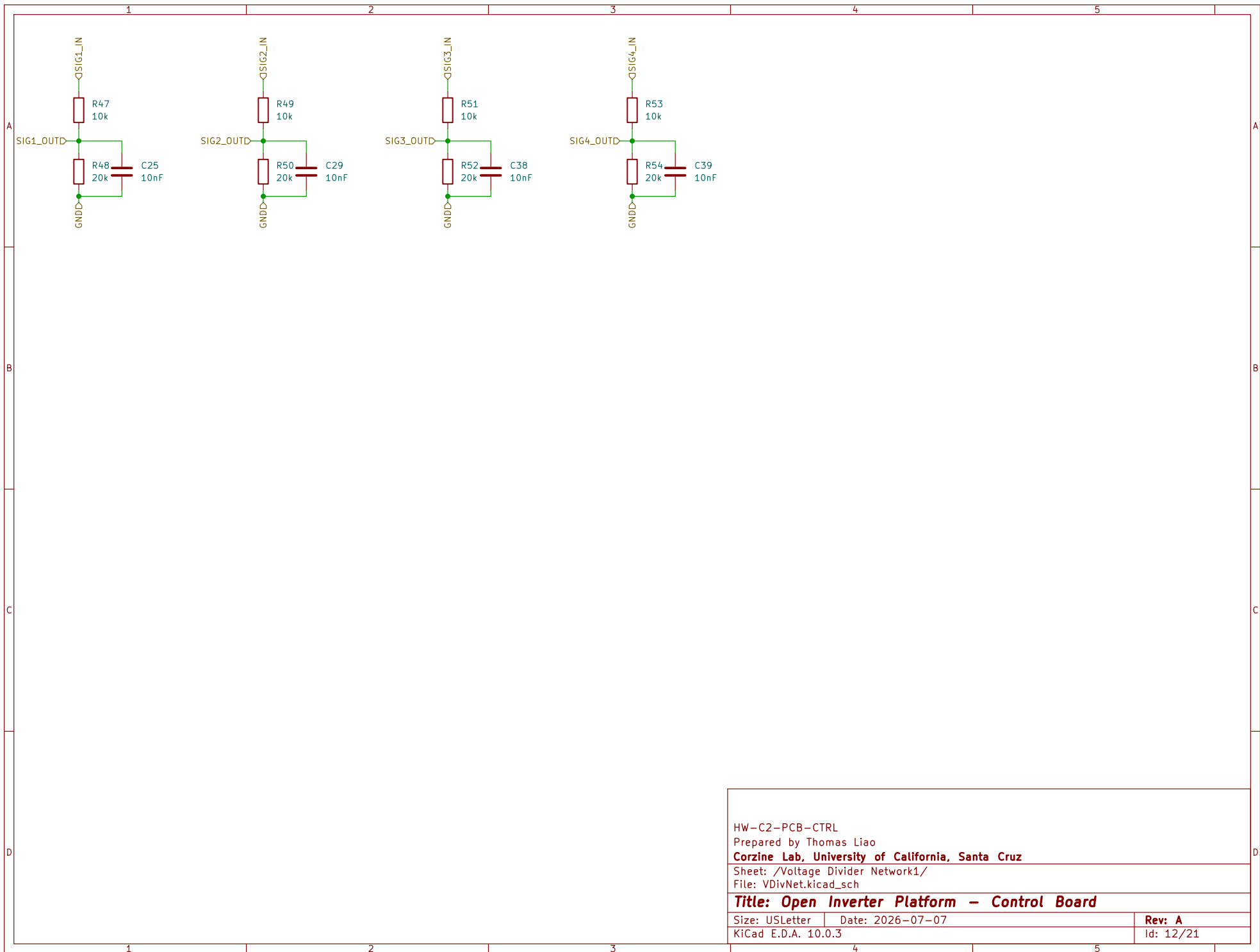












HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

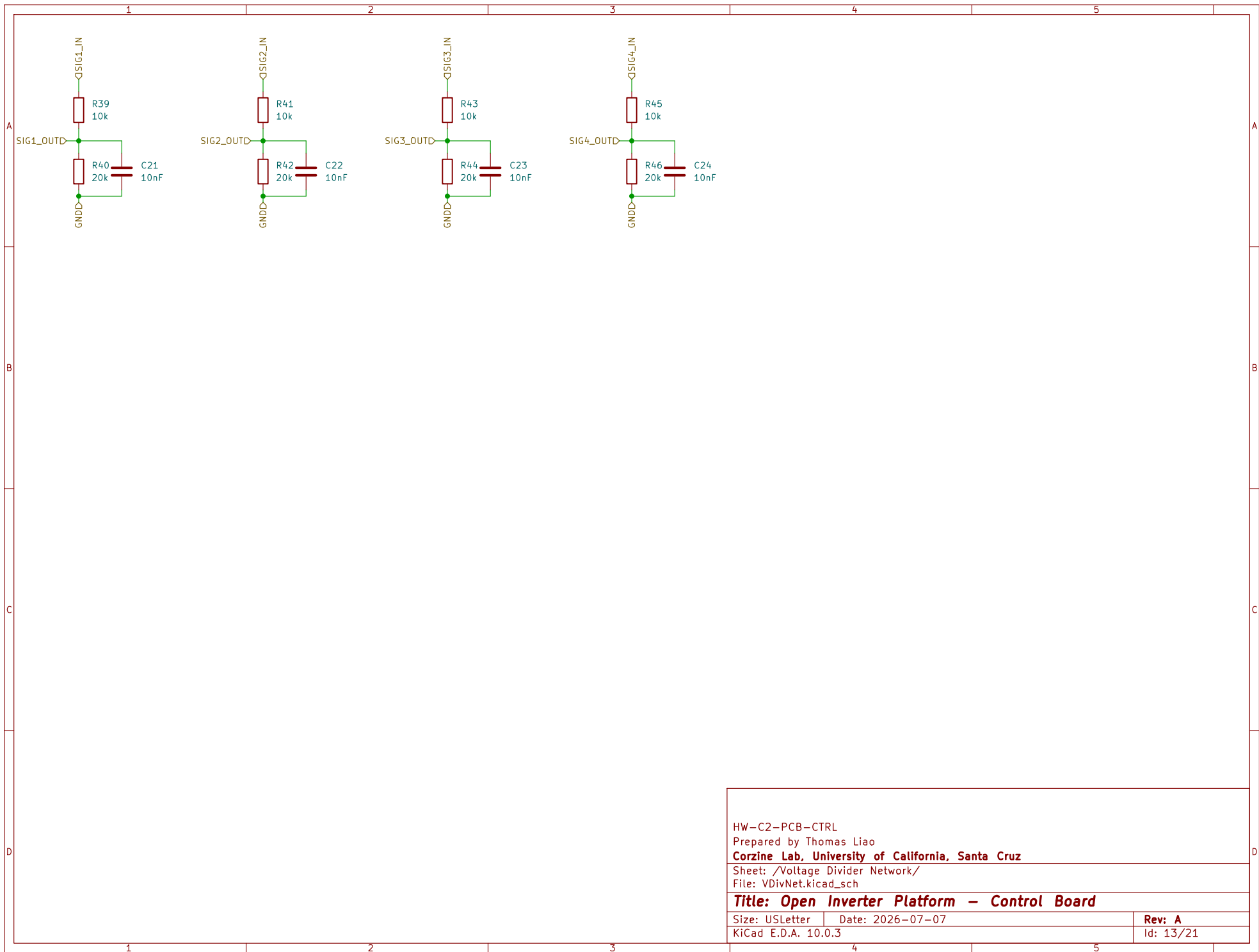
Size: USLetter

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HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

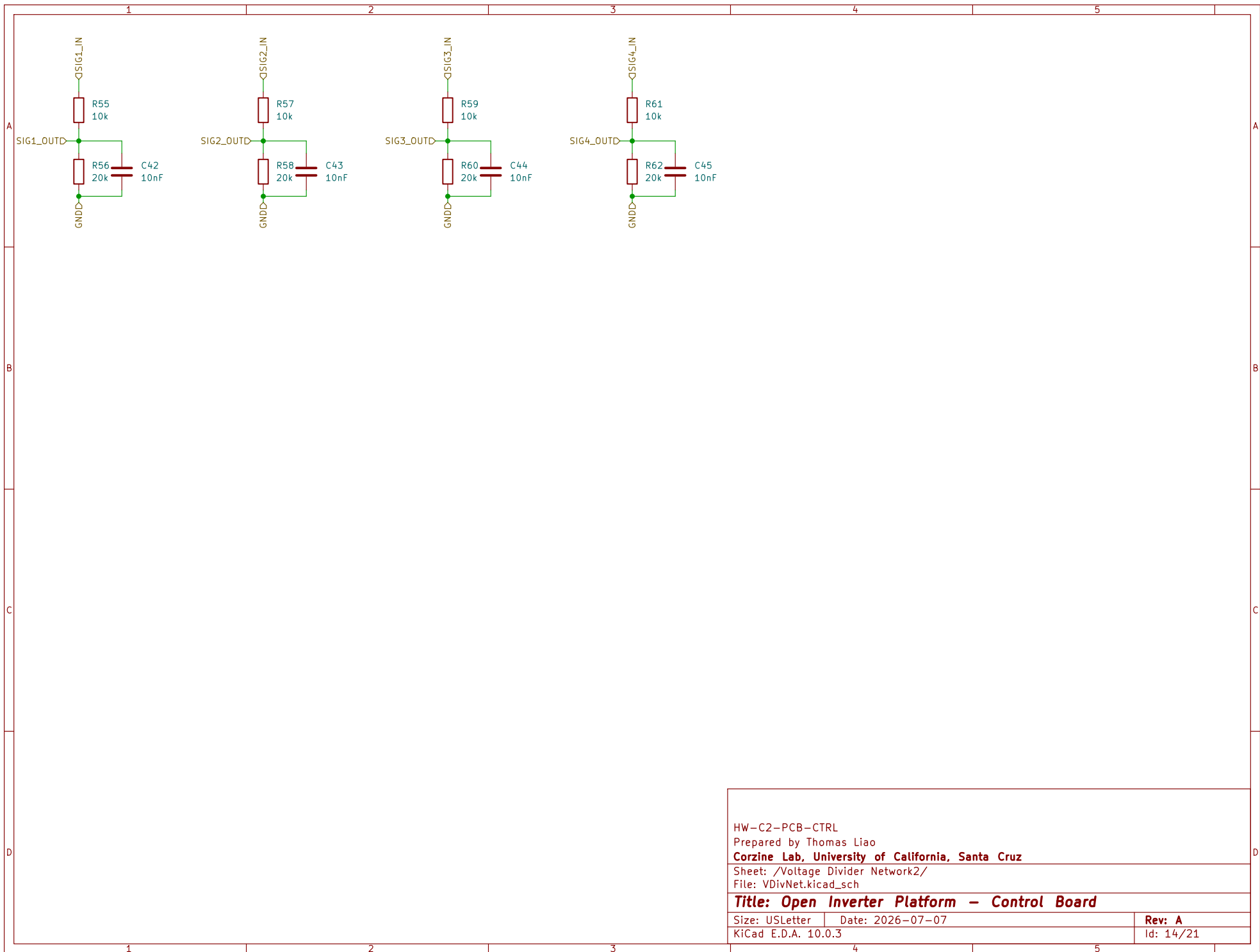
Size: USLetter

Date: 2026-07-07

Rev: A

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Id: 13/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

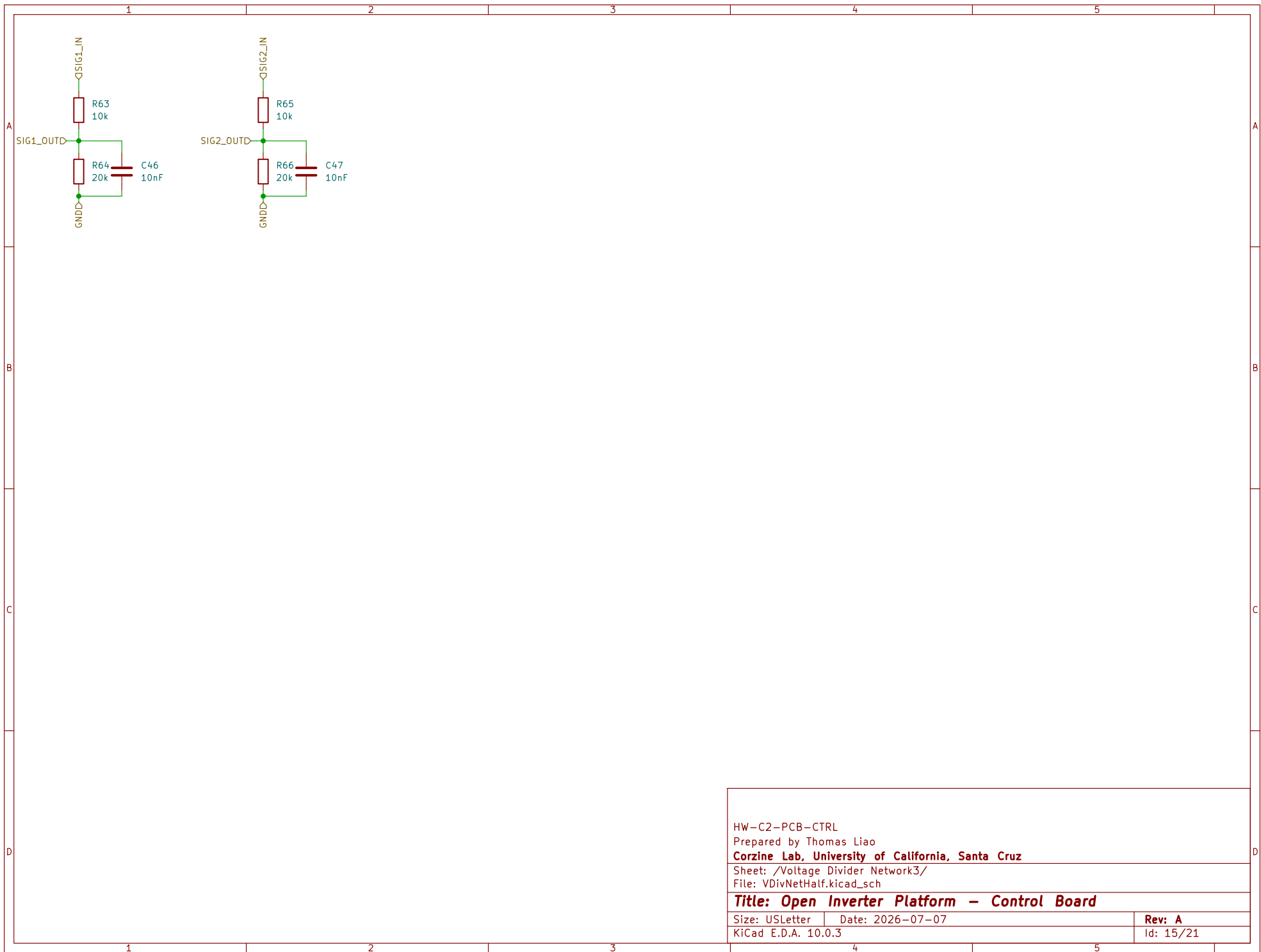
Size: USLetter

Date: 2026-07-07

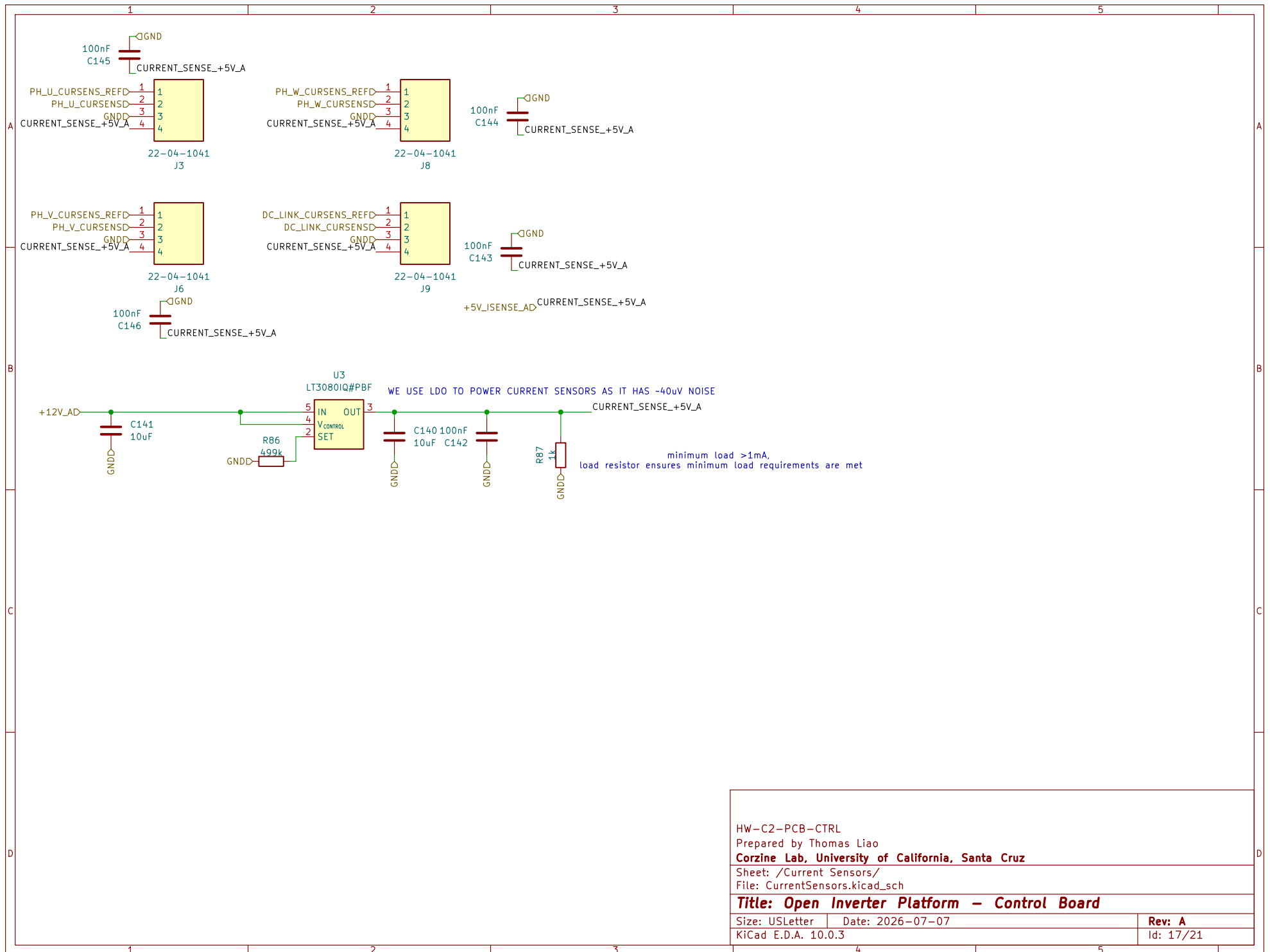
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Id: 14/21



HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Voltage Divider Network3/		
File: VDivNetHalf.kicad_sch		
Title: Open Inverter Platform – Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3	Id: 15/21	



HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter Date: 2026-07-07

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